

Abelian

$ab = ba \rightarrow \text{group} \rightarrow \mathbb{Z}$

$\frac{\langle a, b | ab=ba \rangle}{\mathbb{Z} \times \mathbb{Z}}$

$S^1 \vee S^1$

$T = \mathbb{Z} \times \mathbb{Z}$

$\pi_1(\cdot) = \mathbb{Z} \times \mathbb{Z}$

[illegible]

R/P :
 0-cell: $\{v\}$
 1-cell: $\{x\}$
 2-cell: $\{y\}$

Hand-drawn sketches on grid paper:

- A circle containing a sine wave, with two points on the circle connected by a line segment.
- A flower-like shape with five petals.
- A fish-like shape with a circular body, a tail, and a head.
- An eye-like shape with a circular iris and a pupil.
- A face-like shape with a large nose and a small mouth.
- A square with arrows pointing outwards from each corner.